1. Weather-Based Play Prediction

Build a WEKA application that predicts whether a person should play outdoors based on weather conditions.

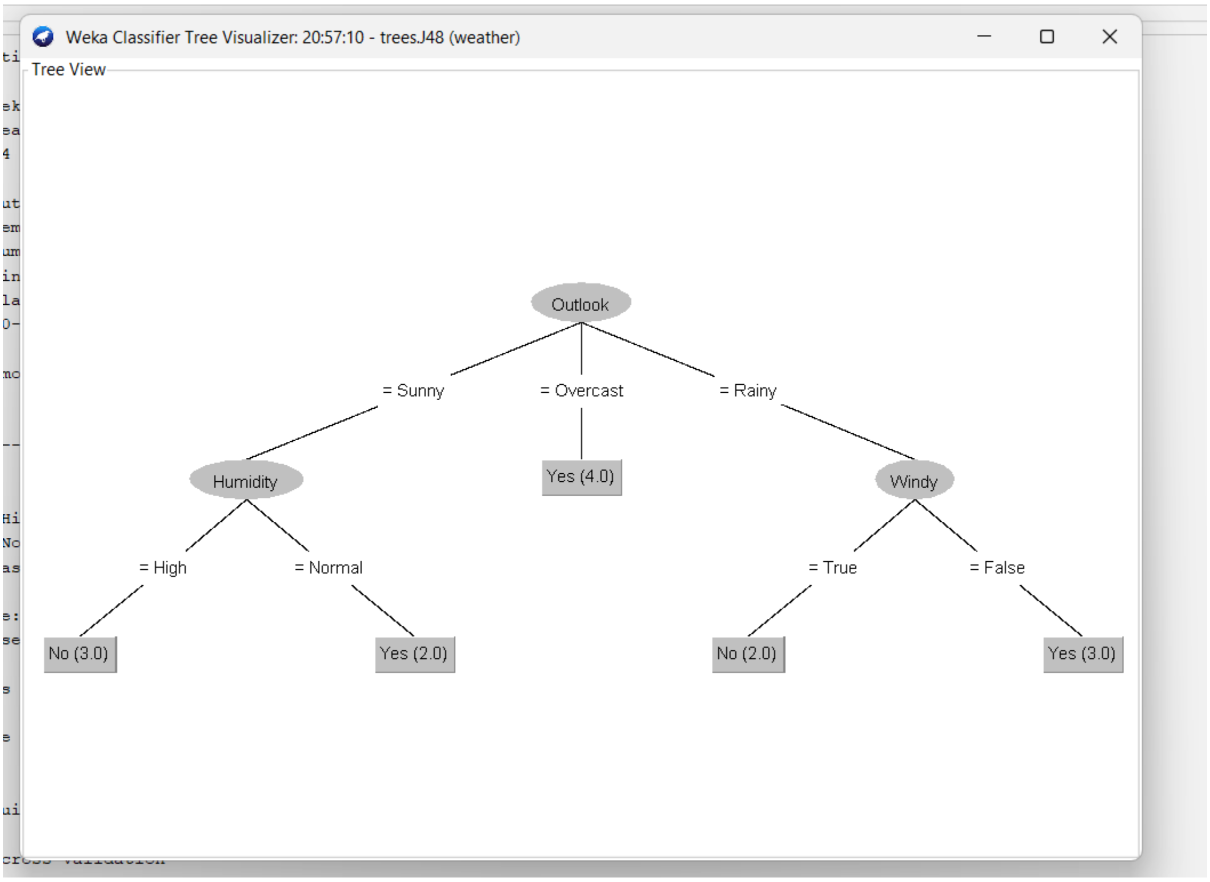
Dataset: Outlook, Temperature, Humidity, Windy

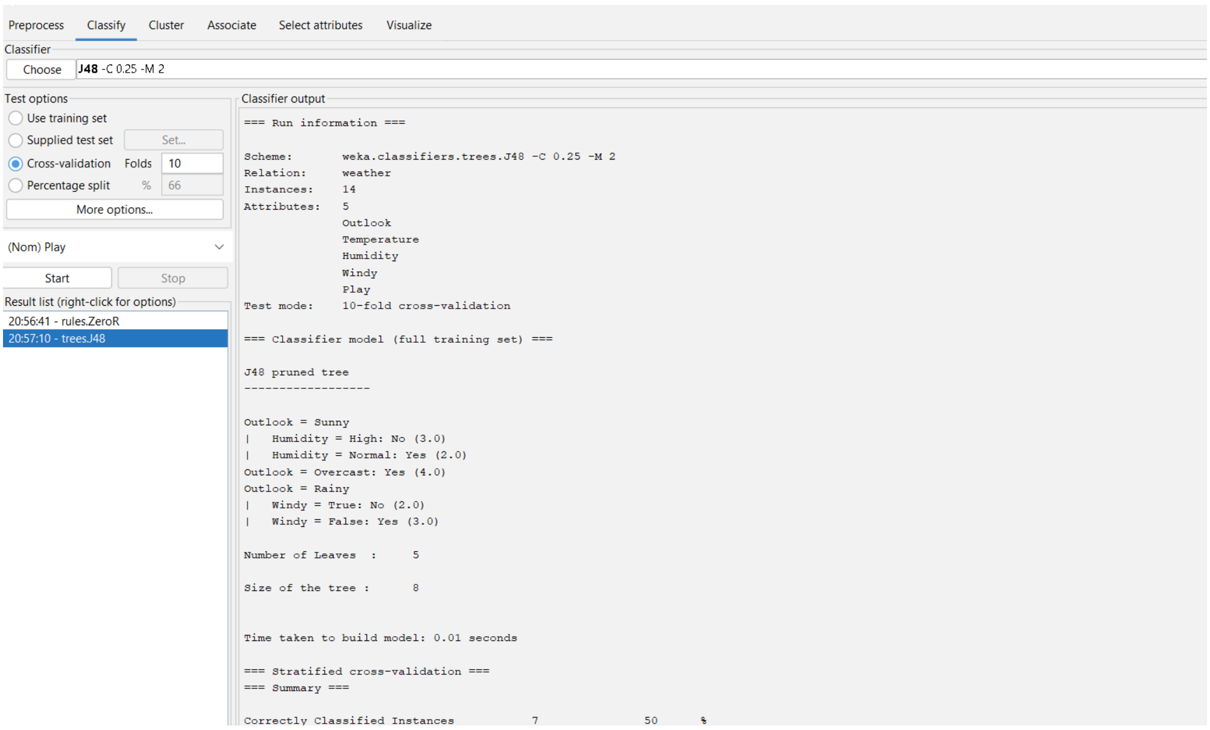
Target: Play (Yes/No)

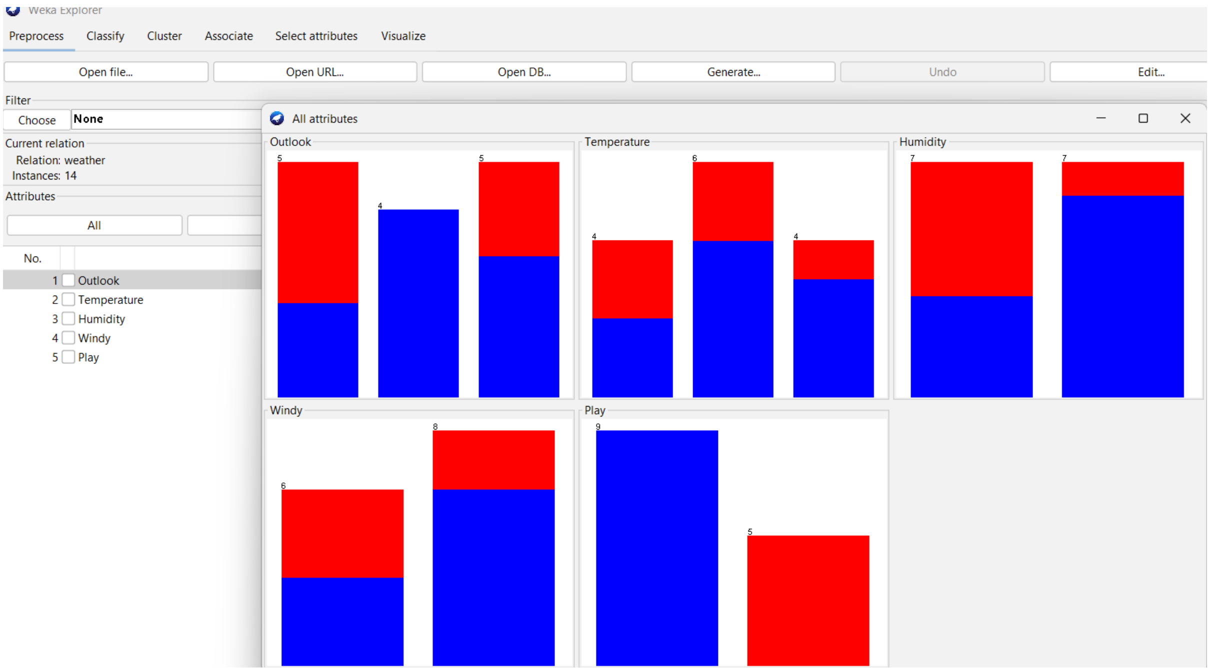
Algorithm: J48 Decision Tree

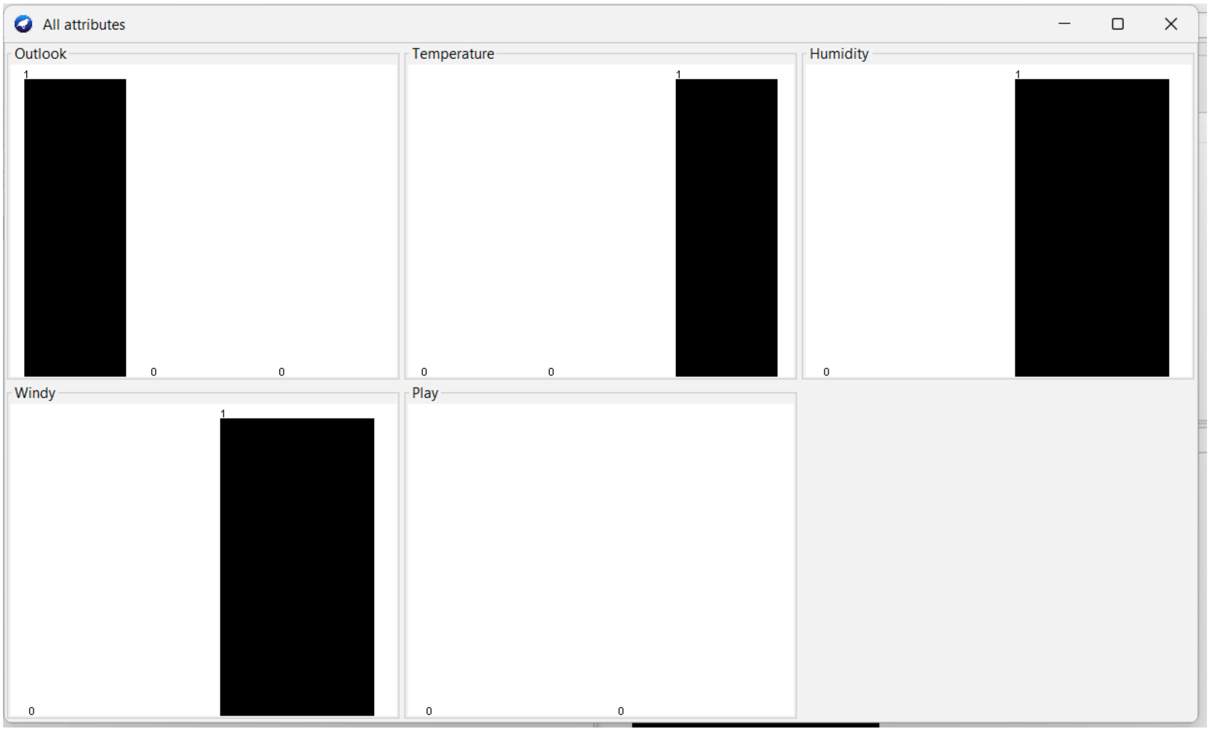
Tasks:

1. Load the dataset into WEKA.
2. Preprocess the data.
3. Apply J48.
4. Visualize the decision tree.
5. Predict Play for a new weather condition.


Weka Classifier Tree Visualizer: 20:57:10 - trees.J48 (weather)
Tree View
Outlook
= Sunny
= Overcast
= Rainy
Humidity
Yes (4.0)
Windy
= High
= Normal
= True
= False
No (3.0)
Yes (2.0)
No (2.0)
Yes (3.0)


Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) Play
Start
Stop
Result list (right-click for options)
20:56:41 - rules.ZeroR
20:57:10 - trees.J48
Classifier output
=== Run information ===
Scheme: weka.classifiers.trees.J48 -C 0.25 -M 2
Relation: weather
Instances: 14
Attributes: 5
Outlook
Temperature
Humidity
Windy
Play
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===
J48 pruned tree
------------------
Outlook = Sunny
| Humidity = High: No (3.0)
| Humidity = Normal: Yes (2.0)
Outlook = Overcast: Yes (4.0)
Outlook = Rainy
| Windy = True: No (2.0)
| Windy = False: Yes (3.0)
Number of Leaves : 5
Size of the tree : 8
Time taken to build model: 0.01 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 7 50 %


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Filter
Choose
None
Current relation
Relation: weather
Instances: 14
Attributes
All
No.
1 Outlook
2 Temperature
3 Humidity
4 Windy
5 Play
All attributes
Outlook
Temperature
Humidity
Windy
Play


All attributes
Outlook
Temperature
Humidity
Windy
Play

2. Flower Species Classification □□

Build a WEKA application that identifies the species of a flower from its physical measurements.

□ Dataset: Iris dataset

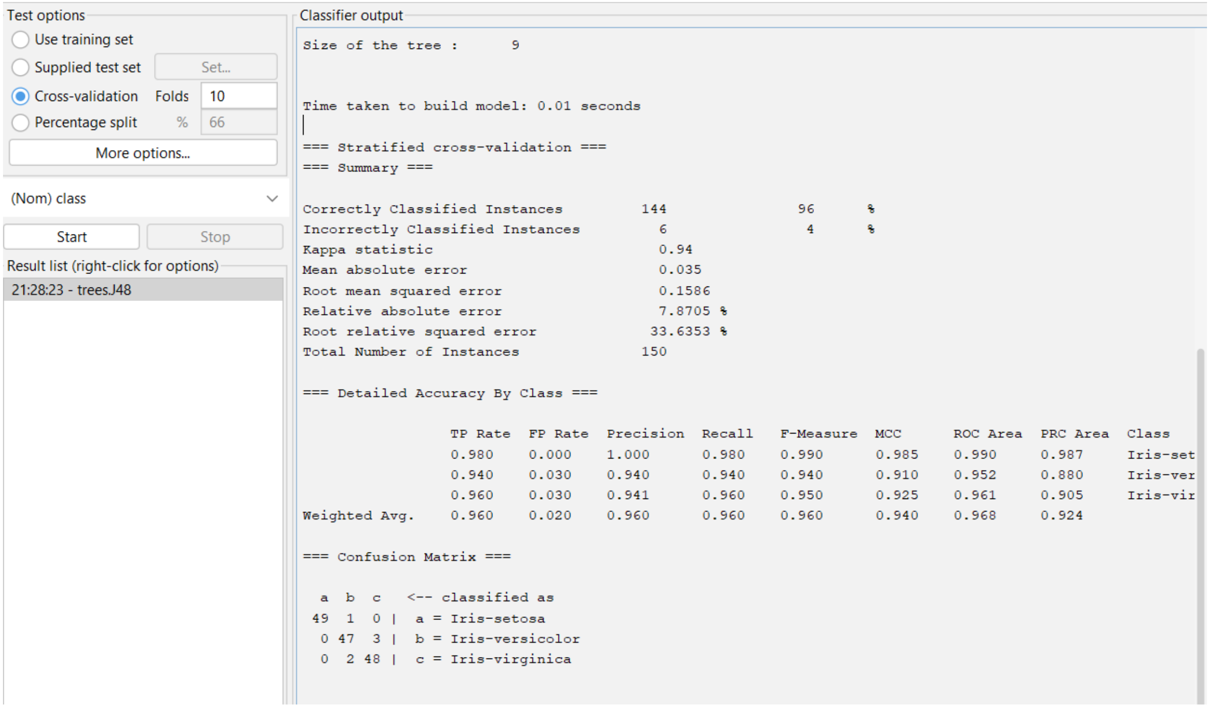
□ Attributes: Sepal Length, Sepal Width, Petal Length, Petal Width

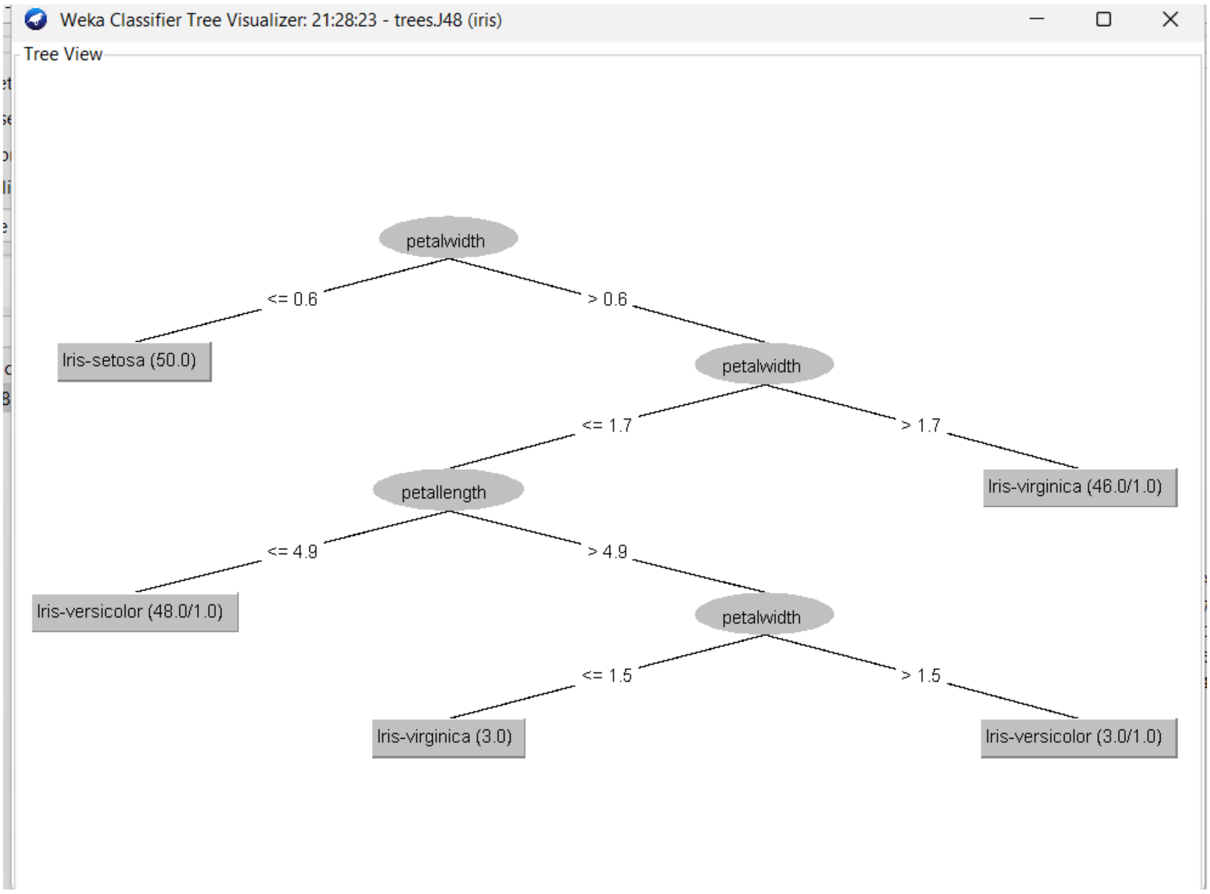
□ Target: Iris Species

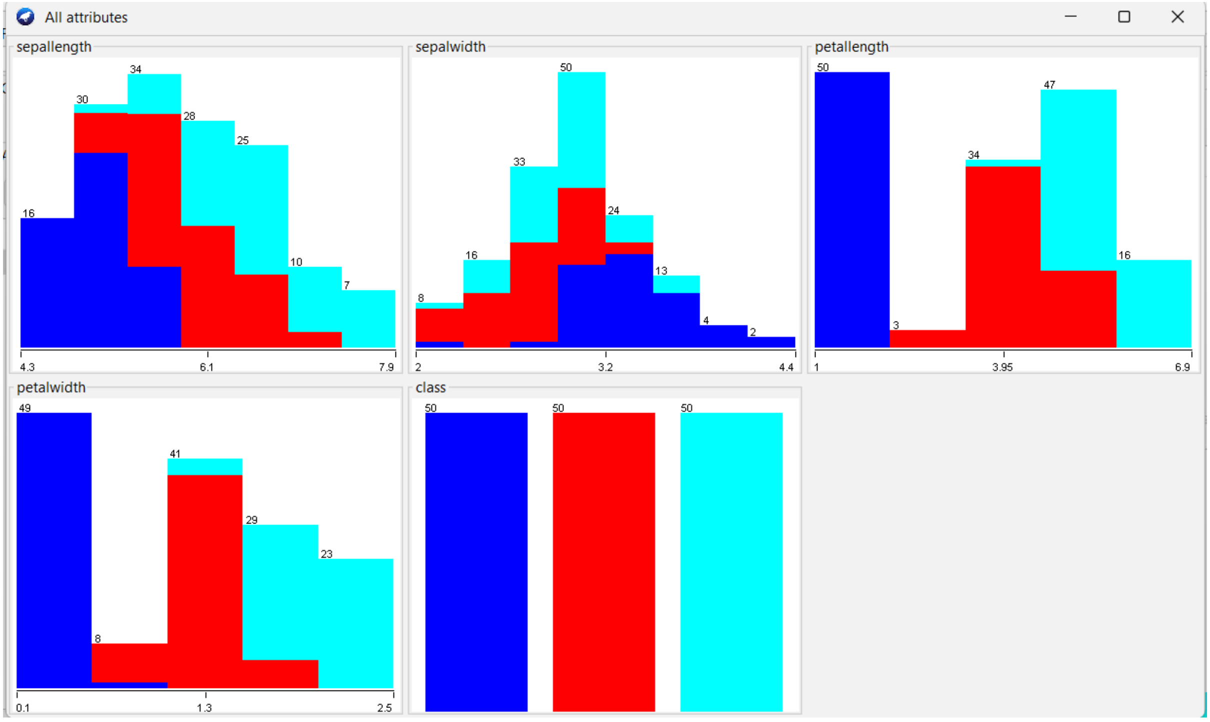
□ Algorithm: J48 or Naive Bayes

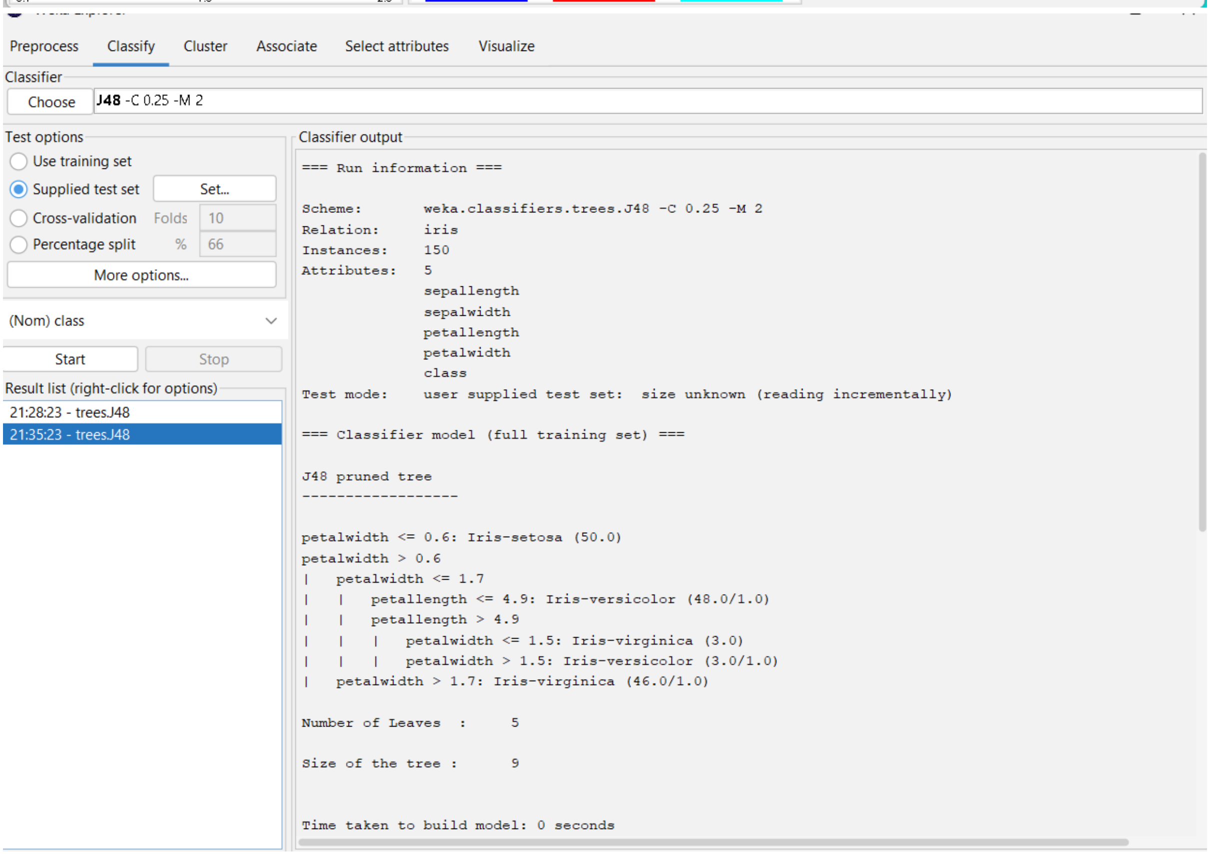
□ Tasks:

1. Load the Iris dataset.
2. Select the classification algorithm.
3. Train the model.
4. Evaluate its accuracy.
5. Classify a new flower.


Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) class
Start
Stop
Result list (right-click for options)
21:28:23 - trees.J48
Classifier output
Size of the tree : 9
Time taken to build model: 0.01 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 144 96 %
Incorrectly Classified Instances 6 4 %
Kappa statistic 0.94
Mean absolute error 0.035
Root mean squared error 0.1586
Relative absolute error 7.8705 %
Root relative squared error 33.6353 %
Total Number of Instances 150
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
0.980 0.000 1.000 0.980 0.990 0.985 0.990 0.987 Iris-set
0.940 0.030 0.940 0.940 0.940 0.910 0.952 0.880 Iris-ver
0.960 0.030 0.941 0.960 0.950 0.925 0.961 0.905 Iris-vir
Weighted Avg. 0.960 0.020 0.960 0.960 0.960 0.940 0.968 0.924
=== Confusion Matrix ===
a b c <-- classified as
49 1 0 | a = Iris-setosa
0 47 3 | b = Iris-versicolor
0 2 48 | c = Iris-virginica


Weka Classifier Tree Visualizer: 21:28:23 - trees.J48 (iris)
Tree View
petalwidth
<= 0.6
> 0.6
Iris-setosa (50.0)
petalwidth
<= 1.7
> 1.7
petallength
Iris-virginica (46.0/1.0)
<= 4.9
> 4.9
Iris-versicolor (48.0/1.0)
petalwidth
<= 1.5
> 1.5
Iris-virginica (3.0)
Iris-versicolor (3.0/1.0)


All attributes
sepallength
34
30
28
25
16
10
7
4.3
6.1
7.9
sepalwidth
50
33
24
16
13
8
4
2
2
3.2
4.4
petallength
50
47
34
16
3
1
3.95
6.9
petalwidth
49
41
29
23
8
0.1
1.3
2.5
class
50
50
50


Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) class
Start
Stop
Result list (right-click for options)
21:28:23 - trees.J48
21:35:23 - trees.J48
Classifier output
=== Run information ===
Scheme: weka.classifiers.trees.J48 -C 0.25 -M 2
Relation: iris
Instances: 150
Attributes: 5
sepallength
sepalwidth
petallength
petalwidth
class
Test mode: user supplied test set: size unknown (reading incrementally)
=== Classifier model (full training set) ===
J48 pruned tree
------------------
petalwidth <= 0.6: Iris-setosa (50.0)
petalwidth > 0.6
| petalwidth <= 1.7
| | petallength <= 4.9: Iris-versicolor (48.0/1.0)
| | petallength > 4.9
| | | petalwidth <= 1.5: Iris-virginica (3.0)
| | | petalwidth > 1.5: Iris-versicolor (3.0/1.0)
| petalwidth > 1.7: Iris-virginica (46.0/1.0)
Number of Leaves : 5
Size of the tree : 9
Time taken to build model: 0 seconds